Discrete Applied Mathematics 321 (2022) 385–391

Contents lists available at ScienceDirect

Discrete Applied Mathematics

journal homepage: www.elsevier.com/locate/dam

Colorings with neighborhood parity condition

Mirko Petruševski [a], Riste Škrekovski [b,*]

[a] Department of Mathematics and Informatics, Faculty of Mechanical Engineering - Skopje, Republic of North Macedonia
[b] Faculty of Mathematics and Physics, University of Ljubljana & Faculty of Information Studies, Novo Mesto, Slovenia

ARTICLE INFO

Article history:
Received 28 December 2021
Received in revised form 1 July 2022
Accepted 25 July 2022
Available online 11 August 2022

Keywords:
Planar graph
Neighborhood
Proper coloring
Odd coloring

ABSTRACT

In this short paper, we introduce a new vertex coloring whose motivation comes from our series on odd edge-colorings of graphs. A proper vertex coloring φ of a graph G is said to be odd if for each non-isolated vertex $x \in V(G)$ there exists a color c such that $\varphi^{-1}(c) \cap N(x)$ is odd-sized. We prove that every simple planar graph admits an odd 9-coloring, and conjecture that 5 colors always suffice.

© 2022 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

All considered graphs in this paper are simple, finite and undirected. We follow [3] for all terminology and notation not defined here. A k-(vertex-)coloring of a graph G is an assignment $\varphi : V(G) \to \{1, \ldots, k\}$. A coloring φ is said to be *proper* if every color class is an independent subset of the vertex set of G. A *hypergraph* $\mathcal{H} = (V(\mathcal{H}), \mathcal{E}(\mathcal{H}))$ is a generalization of a graph, its (hyper-)edges are subsets of $V(\mathcal{H})$ of arbitrary positive size. There are various notions of (vertex-)coloring of hypergraphs, which when restricted to graphs coincide with proper graph coloring. One such notion was introduced by Even at al. [8] (in a geometric setting) in connection with frequency assignment problems for cellular networks, as follows. A coloring of a hypergraph $\mathcal{H}$ is *conflict-free (CF)* if for every edge $e \in \mathcal{E}(\mathcal{H})$ there is a color c that occurs exactly once on the vertices of e. The *CF chromatic number* of $\mathcal{H}$ is the minimum k for which $\mathcal{H}$ admits a CF k-coloring. For graphs, Cheilaris [5] studied the *CF coloring with respect to neighborhoods*, that is, the coloring in which for every non-isolated vertex x there is a color that occurs exactly once in the (open) neighborhood $N(x)$, and proved the upper bound $2\sqrt{n}$ for the CF chromatic number of a graph of order n. For more on CF colorings see, e.g., [7,10,12,15,20].

A similar but considerably less studied notion (concerning a weaker requirement for the occurrence of a color) was introduced by Cheilaris et al. [6] as follows. An *odd coloring* of a hypergraph $\mathcal{H}$ is a coloring such that for every edge $e \in \mathcal{E}(\mathcal{H})$ there is a color c with an odd number of vertices of e colored by c. Particular features of the same notion (under the name *weak-parity coloring*) have been considered by Fabrici and Göring [9] (in regard to face-hypergraphs of planar graphs) and also by Bunde et al. [4] (in regard to coloring of graphs with respect to paths, i.e., path-hypergraphs). For various edge colorings of graphs with parity condition required at the vertices we refer the reader to [2,11,13,14,16–18].

In this paper we study certain aspects of odd colorings for graphs with respect to (open) neighborhoods, that is, the colorings of a graph G such that for every non-isolated vertex x there is a color that occurs an odd number of times in the neighborhood $N_G(x)$. Our focus is on colorings that are at the same time proper, and we mainly confine to planar graphs.

Let us denote by $\chi_o(G)$ the minimum number of colors in any proper coloring of a given graph G that is odd with respect to neighborhoods, call this the *odd chromatic number* of G. Note that the obvious inequality $\chi(G) \leq \chi_o(G)$ may be

* Corresponding author.
E-mail addresses: mirko.petrushevski@gmail.com (M. Petruševski), skrekovski@gmail.com (R. Škrekovski).

https://doi.org/10.1016/j.dam.2022.07.018
0166-218X/© 2022 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

strict; e.g., $\chi(C_4) = 2$ whereas $\chi_o(C_4) = 4$. Similarly, $\chi(C_5) = 3$ whereas $\chi_o(C_5) = 5$. In fact, the difference $\chi_o(G) - \chi(G)$ can acquire arbitrarily large values. Indeed, let G be obtained from K_n ($n \geq 2$) by subdividing each edge once. Since G is bipartite, $\chi(G) = 2$. On the other hand, it is readily seen that $\chi_o(G) \geq n$. Note in passing another distinction between the chromatic number and the odd chromatic number. The former graph parameter is monotonic in regard to the 'subgraph relation', that is, if $H \subseteq G$ then $\chi(H) \leq \chi(G)$. This nice monotonicity feature does not hold for the odd chromatic number in general. For example, C_4 is a subgraph of the kite $K_4 - e$, but nevertheless we have $\chi_o(C_4) = 4 > 3 = \chi_o(K_4 - e)$.

The Four Color Theorem [1,19] asserts the tight upper bound $\chi(G) \leq 4$ for the chromatic number of any planar graph G. One naturally starts wondering about an analogous bound for the odd chromatic number of all planar graphs. Since $\chi_o(C_5) = 5$, four colors no longer suffice. It is our belief that five colors always suffice.

Conjecture 1.1. *For every planar graph G it holds that $\chi_o(G) \leq 5$.*

The bound in Conjecture 1.1 is achieved for infinitely many planar graphs. Namely, let $\mathcal{F} = \{G : G$ is a non-trivial connected graph such that every block of G is isomorphic to $C_5\}$.

Proposition 1.2. *If $G \in \mathcal{F}$ then $\chi_o(G) = 5$.*

Proof. Consider a counter-example G with minimum number of blocks, $b(G)$. Since $G \neq C_5$, we have that $b(G) \geq 2$. Let B be an endblock of G, and let v_1, v_2, v_3, v_4, v_5 be the vertices of the 5-cycle B met on a circular traversing that starts at the unique cut vertex v_1 contained within B. Consider the graph $G' = G - \{v_2, v_3, v_4, v_5\}$. Since $G' \in \mathcal{F}$ and $b(G') = b(G) - 1$, it holds that $\chi_o(G') = 5$. Take an odd coloring of G' with color set $\{1, 2, 3, 4, 5\}$ such that v_1 is colored by 1 and the color 4 has odd number of occurrences in $N_{G'}(v_1)$. Extend to $V(G)$ by assigning the color i to v_i, for $i = 2, 3, 4, 5$. This clearly gives an odd 5-coloring of G. Consequently, since we are supposing $\chi_o(G) \neq 5$, there must exist an odd coloring c of G with color set $\{1, 2, 3, 4\}$. Upon permutation of colors, we may assume that $c(v_i) = i$ for $1 \leq i \leq 4$. Hence, it must be that $c(v_5) = 2$. Thus, any color which satisfies the odd-condition in $N_G(v_1)$ has an odd number of occurrences in $N_{G'}(v_1)$ as well. It follows that the restriction of c to G' is an odd 4-coloring, contradicting the minimality choice of G. □

The main purpose of this paper is to provide the first support to Conjecture 1.1 by proving the following.

Theorem 1.3. *For every planar graph G it holds that $\chi_o(G) \leq 9$.*

The rest of the article is organized as follows. In the next section we provide a proof of Theorem 1.3. And in Section 3 we briefly convey some of our ideas for possible further work on the topic of proper colorings of graphs that are odd with respect to neighborhoods.

2. Proof of Theorem 1.3

Throughout, we will refer to a coloring of this kind as a *nice coloring*; that is, a nice coloring is a (proper) odd coloring which uses at most 9 colors. Arguing by contradiction, let G be a counter-example of minimum order $n = n(G)$. Clearly, G is connected and has $n \geq 10$ vertices. We proceed to exhibit several structural constraints of G.

Claim 1. *The minimum degree $\delta(G)$ equals 5.*

Since G is a connected planar simple graph with $n > 2$ vertices, the inequality $\delta(G) \leq 5$ is a consequence of Euler's formula. Consider a vertex v of degree $d_G(v) = \delta(G)$. If $d_G(v) = 1$ or 3, then take a nice coloring of $G - v$. By forbidding at most six colors at v (namely, at most three colors used for v's neighbors and at most three additional colors in regard to oddness concerning the neighborhoods in $G - v$ of these vertices), the coloring extends to a nice coloring of G.

Suppose next that $d_G(v) = 2$, and say $N_G(v) = \{x, y\}$. Construct G' by removing v from G plus connecting x and y if they are not already adjacent. By minimality, G' admits a nice coloring c. Say $c(x) = 1$ and $c(y) = 2$, and let color(s) $1'$ and $2'$, respectively, have odd number of occurrences in $N_{G'}(x)$ and $N_{G'}(y)$. If there are several possibilities for $1'$, then choose $1' \neq 2$; do similarly for $2'$ in regard to 1. Extend the coloring to G by using for v a color different from $1, 2, 1', 2'$. The properness of the coloring is clearly preserved. As for the weak-oddness concerning neighborhoods, v is fine because $1 \neq 2$. If $1' \neq 2$ then x is fine since $1'$ remains to be odd on $N_G(x)$. Contrarily, if $1' = 2$ then $c(v)$ is odd on $N_G(x)$. Similarly, the vertex y is also fine.

Finally, suppose $d_G(v) = 4$ and let $w \in N_G(v)$. Remove v and connect w by an edge to every other non-adjacent neighbor of v. The obtained graph G'' is simple and planar. Indeed, it can be equivalently obtained from G by contracting the edge vw (if parallel edges arise through possible mutual neighbors of v and w, then for each such adjacency a single edge is kept and its copies are deleted). Notice that under any nice coloring of G'', the color of w occurs exactly once in $N_G(v)$. Therefore, any such coloring extends to a nice coloring of G by forbidding at most eight colors at v.

We conclude that $d_G(v) = 5$. □

We refer to any vertex of degree d as a *d-vertex*. Similarly, a vertex of degree at least d is a *d^+-vertex*. Analogous terminology applies to faces in regard to a planar embedding of G.

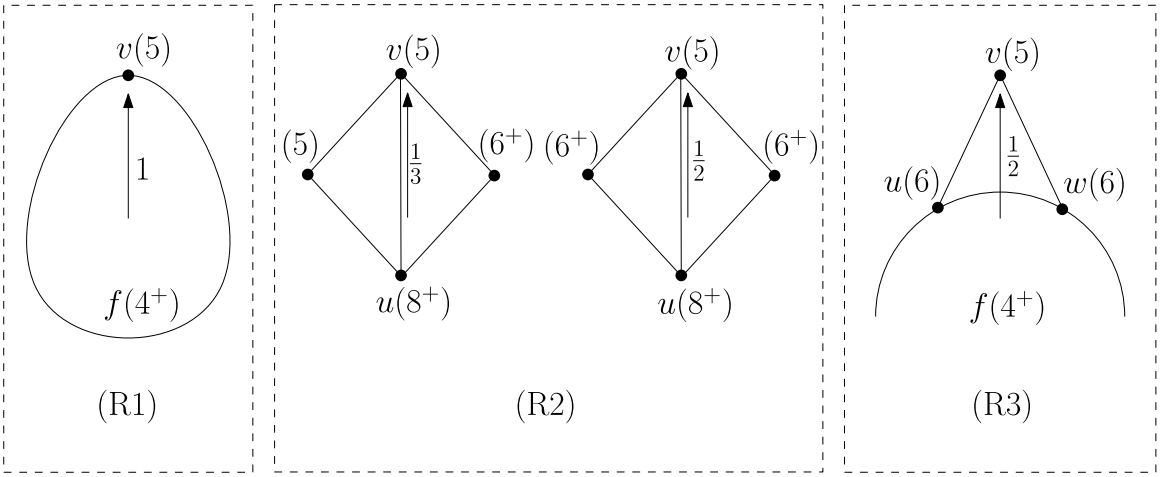

Fig. 1. In (R2), the numbers standing in brackets beside vertices or faces indicate their degrees. In (R3), f is not a 4-face incident with two 5-vertices and uvw is an adjacent triangular face.

Claim 2. *If v is a 5-vertex of G, then it has at most one neighbor of odd degree.*

Suppose each of two vertices $u, w \in N_G(v)$ has an odd degree in G. Consider a nice coloring c of $G - v$. We intend to extend c to G. Since $N_G(u)$ and $N_G(w)$ are odd-sized, no color is blocked at v in regard to oddness in the neighborhoods of u and w. Moreover, as $N_G(v)$ is odd-sized as well, the oddness of a color in this particular neighborhood is guaranteed. Therefore, by forbidding at most eight colors at v (all of $c(N_G(v))$ and at most 3 additional colors concerning oddness in neighborhoods of the vertices forming $N_G(v)\backslash\{u, w\}$), the coloring c extends to a nice coloring of G. □

Since the graph G is connected, for an arbitrary planar embedding Euler's formula gives

$$|V(G)| - |E(G)| + |F(G)| = 2,$$

where $F(G)$ is the set of faces. Our proof is based on the discharging technique. We assign initial charges to the vertices and faces according to the left-hand side of the following equality (which immediately follows from Euler's formula):

$$\sum_{v\in V(G)} (d(v) - 6) + \sum_{f\in F(G)} (2d(f) - 6) = -12.$$

Thus, any vertex v receives charge $d(v) - 6$, and any face f obtains charge $2d(f) - 6$. As G is simple, for every face f it holds that $d(f) \geq 3$, implying that its initial charge is non-negative. By Claim 1, the only vertices that have negative initial charge are the 5-vertices of G (each is assigned with charge -1).

Claim 3. *Any d-face $f \in F(G)$ is incident with at most $\lfloor \frac{2d}{3} \rfloor$ 5-vertices.*

By Claim 2, on a facial walk of f no three consecutive vertices are of degree 5. From this, the stated upper bound for the number of 5-vertices incident with f follows immediately. □

We use the following discharging rules (cf. Fig. 1):

(R1) Every 4^+-face f sends charge 1 to any incident 5-vertex v.
(R2) Every 8^+-vertex u sends charge to any adjacent 5-vertex v if both faces incident with the edge uv are triangles, as follows: if one of these triangles has another 5-vertex (besides v), then the sent charge equals $\frac{1}{3}$; otherwise, the sent charge equals $\frac{1}{2}$. (In view of Claim 2, it cannot be that both these triangles have a 5-vertex $\neq v$.)
(R3) Every 4^+-face f sends through every incident edge uw with $d(u) = d(w) = 6$ charge $\frac{1}{2}$ to a 5-vertex v if uvw is a triangular face and f is not a 4-face incident with two 5-vertices.

Claim 4. *No 8^+-vertex becomes negatively charged by applying (R2).*

Suppose there is a d-vertex u with $d \geq 8$ that becomes negatively charged by applying Rule 2. Recall that its initial charge was $d - 6$. Consider a circular ordering of $N_G(u)$ in regard to the embedding of G. Note that no three consecutive neighbors $v_i, v_{i+1}, v_{i+2} \in N_G(u)$ have received charge from u during the discharging process, for otherwise v_{i+1} contradicts Claim 2. Therefore, u gave charge to at most $\lfloor \frac{2d}{3} \rfloor$ of its neighbors, and at most $\frac{1}{2}$ of charge per neighbor. Consequently, $d - 6 < \frac{2d}{3} \cdot \frac{1}{2}$. Equivalently, $d < 9$. So u is an 8-vertex. Moreover, since we are supposing that it is negatively charged after applying Rule 2, the vertex v gave charge to precisely five 5-vertices. However, then it is easily seen that the initial charge 2 of v is reduced by at most $\frac{11}{6} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{2}$, a contradiction. □

Claim 5. *No 4^+-face becomes negatively charged by applying (R1) or (R3).*

Suppose there is a d-face f that becomes negatively charged by applying Rules 1 or 3. Recall that its initial charge was $2d-6$. For the purposes of this proof, it is useful to think of the charge sent by f according to Rule 3 as follows. Consider any maximal sequence of 'consecutive' triangular faces that are adjacent to f, and split the sequence into 2-tuples and a possible 1-tuple. For every such 2-tuple of triangular faces, the total charge $1=\frac{1}{2}+\frac{1}{2}$ first goes to their common 6-vertex, and only afterwards splits evenly and reaches the targeted 5-vertices (possibly the same). And for every such 1-tuple the charge $\frac{1}{2}$ coming from f to this face first splits evenly and goes to the pair of 6-vertices shared with f, and only afterwards reaches the targeted 5-vertex.

With this perspective in mind, consider a facial walk of f and notice that during the discharging it gives away to each incident vertex charge of either 0, $\frac{1}{4}$, or 1; moreover, no three consecutive vertices receive from f charge 1 (according to Claim 2). Hence, on at most $\lfloor\frac{2d}{3}\rfloor$ occasions f gives charge 1, and on the other $\lfloor\frac{d+1}{3}\rfloor$ occasions f gives charge at most $\frac{1}{4}$. Consequently, $2d-6<\frac{2d}{3}+\frac{d+1}{12}$. Equivalently, $d<\frac{73}{15}$. So f is a 4-face. However, it is easily seen that a 4-face gives away at most 2 of its initial charge (which also equals 2), a contradiction. □

Since the total charge remains negative (it equals -12), from Claims 4 and 5 it follows that there is a 5-vertex v which remains negatively charged even after applying the discharging rules. By (R1) and (R2), we have the following.

Claim 6. *The 5-vertex v has only 3-faces around it, and it is a neighbor of at most two 8^+-vertices. Moreover, if v neighbors exactly two 8^+-vertices, then these three vertices have another 5-vertex as a common neighbor.*

Let v_1, v_2, v_3, v_4, v_5 be the neighbors of v in a circular order regarding the considered plane embedding, i.e., such that vv_1v_2, vv_2v_3, vv_3v_4, vv_4v_5, vv_5v_1 are the five 3-faces incident with v.

Claim 7. *For some $j=1,2,\ldots,5$, the vertices v_j, v_{j+2} are not adjacent and their only common neighbors are v, v_{j+1}.*

Arguing by contradiction, suppose that every pair of vertices v_j, v_{j+2} are either adjacent or have a common neighbor $\neq v, v_{j+1}$. This readily implies the existence of a vertex $x\neq v$ that is adjacent to all five vertices v_1, v_2, v_3, v_4, v_5. Now take a nice coloring c of $G-v$, and let $s=|c(N_G(v))|$. In view of the 5-cycle $v_1v_2v_3v_4v_5$, it holds that $s\in\{3,4,5\}$. In case $s=3$, the coloring c extends to a nice coloring of G since at most 8 colors are forbidden at v. These are the three colors used on $N_G(v)$ and at most five additional colors in regard to 'oddness' in $N_{G-v}(v_i)$ for $i=1,2,\ldots,5$. Hence $s\in\{4,5\}$, and from this we are able to further deduce that there are precisely s vertices $v_i (i=1,2,\ldots,5)$ such that $c(v_{i-1})\neq c(v_{i+1})$.

Unless the coloring c extends to G, each of the nine available colors is blocked at the vertex v, either due to properness or to 'oddness'. Consequently, there are at least $(9-s)$ vertices among the v_i's each of which blocks at v a separate color $\neq c(v_1), c(v_2), c(v_3), c(v_4), c(v_5)$ in regard to 'oddness' in its neighborhood within $G-v$. Clearly, any such v_i is of even degree in G and any color from $\{c(v_{i-1}), c(v_{i+1})\}$ appears an even number of times in $N_{G-v}(v_i)$. Moreover, for at least four such vertices v_i, say v_1, v_2, v_3, v_4, it also holds that $c(v_{i-1})\neq c(v_{i+1})$. Noting that the color $c(x)$ occurs in every of the four neighborhoods $N_{G-v}(v_i)$ for $i=1,\ldots,4$, it follows that $c(x)$ has an odd number of appearances in such a neighborhood for at most one vertex v_i. Therefore, at least three v_i's are of degree $\geq 2+2+2+1=7$ in $G-v$. (Namely, each of the colors $c(v_{i-1}), c(v_{i+1}), c(x)$ yields a summand of 2, and the summand of 1 refers to the color blocked by v_i due to 'oddness' in $N_{G-v}(v_i)$). However, this yields at least three 8^+-neighbors of v in G, contradicting Claim 6. □

Let v_j, v_{j+2} fulfill Claim 7. Consider the graph $(G-v)/\{v_j, v_{j+2}\}$ obtained from $G-v$ by identifying v_j and v_{j+2}, and then deleting one of the arising two parallel edges (between the new vertex $\{v_j, v_{j+2}\}$ and the vertex v_{j+1}). Take a nice coloring of $(G-v)/\{v_j, v_{j+2}\}$, and look into the inherited coloring c of $G-v$.

Claim 8. *All v_i's are of even degree in G, and the coloring c is nice.*

Notice that besides the odd-condition at v_{j+1}, the coloring c is nice on $G-v$. If $d_G(v_{j+1})$ is odd, then when adding v to $G-v$ we do not need to care about odd-condition at v_{j+1}, as it will automatically hold. But then we need to avoid only 8 colors when coloring v, which is doable, a contradiction. Hence $d_G(v_{j+1})$ is even. Consequently, c is a nice coloring of $G-v$ (as $d_{G-v}(v_{j+1})$ is odd and the rest of the conditions are fine). Since c does not extend to a nice coloring of G, by reasoning in the same manner as at the beginning of the proof of Claim 7, we are able to conclude that $|c(N_G(v))|\geq 4$. On the other hand, by the construction of c we clearly have $|c(N_G(v))|<5$. So it must be that exactly four colors appear in $N_G(v)$. Moreover, as c is non-extendable, each of the vertices v_1, v_2, v_3, v_4, v_5 blocks at v a separate color $\notin c(N_G(v))$ in regard to already (and uniquely) fulfilled 'oddness' in its neighborhood in $G-v$. However, this readily implies that each v_i is of even degree in G. □

The proof of Claim 8 certifies that, upon permuting colors, the following color distribution occurs under c: the colors $1,2,1,3,4$ are used on v_1, v_2, v_3, v_4, v_5, respectively, and each of the colors $5,6,7,8,9$ happens to be the only color with an odd number of appearances on a separate neighborhood $N_{G-v}(v_i)$ (cf. Fig. 2(a)). Moreover, as colors $5,6,7,8,9$ are 'exclusive' in regard to 'oddness' on $N_{G-v}(v_1), N_{G-v}(v_2), N_{G-v}(v_3), N_{G-v}(v_4), N_{G-v}(v_5)$, respectively, any of the colors $1,2,3,4$ has an even (possibly 0) number of occurrences in each set $N_{G-v}(v_i)$ $(i=1,\ldots,5)$. Consequently, the situation depicted in Fig. 2(b) is present, that is, at four of the v_i's (namely, v_1, v_2, v_4, v_5) we are guaranteed two more 'free hanging' colors per vertex. Let f_i $(i=1,\ldots,5)$ be the face incident with v_iv_{i+1} but not with v (cf. Fig. 2(b)). Note that f_1, f_2, f_3, f_4, f_5 need not be pairwise distinct. We say that f_i is *convenient* if it is a 4^+-face and v_i, v_{i+1} are 6-vertices whose only common neighbor is v.

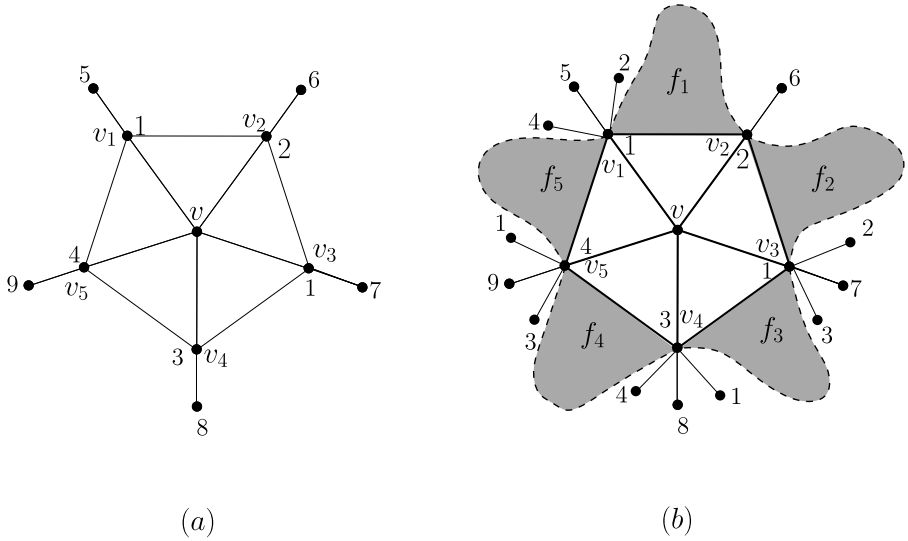

Fig. 2. Local color distribution, guaranteed 'free hanging' colors and the faces adjacent with the local triangulation around v.

Claim 9. *If all of v_1, v_2, v_3, v_4, v_5 are 6-vertices in G, then at least two members of the list f_1, f_2, f_3, f_4, f_5 are convenient.*

It suffices to observe the following: if a given f_k is a 3-face, then the vertices v_k and v_{k+1} must share a free hanging color. Since all the v_i's are 6-vertices, it follows that v_3 and v_4 do not have a common free hanging color. Consequently, f_3 is convenient. Quite similarly, f_5 is convenient. □

Following the same line of reasoning, we also conclude the following:

Claim 10. *If precisely one of v_1, v_2, v_3, v_4, v_5 is an 8^+-vertex in G, then some f_i is convenient.*

Indeed, we may assume that neither v_1 nor v_5 is the 8^+-neighbor of v. Then v_1, v_5 are 6-vertices without a common free hanging color. Consequently, f_5 is convenient. □

Our next (and final) claim assures that every convenient face f_i sends charge $\frac{1}{2}$ to v, in accordance with (R3).

Claim 11. *No convenient face is a quadrangle $ABCD$ where A, B are 5-vertices and C, D are 6-vertices.*

Arguing by contradiction, we suppose that the situation depicted in Fig. 3(a) is present. Without loss of generality, we may assume that B and D are non-adjacent and $N_G(B) \cap N_G(D) = \{A, C\}$. Indeed, if each of the pairs of vertices A, C and B, D are either adjacent or share a common neighbor outside the set $\{A, B, C, D\}$, then by planarity, the situation depicted in Fig. 3(b) occurs. However, then C and D have a common neighbor $\neq v$, which contradicts that the considered face is convenient.

With our assumption for the pair B, D, we look at the graph $G' = (G - \{v, A, C\})/\{B, D\}$, that is, G' is obtained from $G - \{v, A, C\}$ by identifying B and D. Since G' is planar and smaller than G, it admits a nice coloring. Consider the inherited partial coloring c of G. Note that vertices B and D are colored the same under c, and the only uncolored vertices are v, A and C. We extend c to a nice coloring of G as follows.

First we color v, and for the time being we do not care about preserving oddness in $N_{G-A-C}(D)$. Apart from the color $c(D)$ and possibly a second color in regard to unique oddness in $N_{G-\{v,A\}}(C)$, at most $6 = 3 \cdot 2$ more colors are forbidden at v (by its three neighbors $\neq C, D$ and oddness in their respective neighborhoods within $G - \{v, A, C\}$). Thus there is a color which is available for v.

Next we color C. There are at most $8 = 1 + 1 + 3 \cdot 2$ colors forbidden at C (namely, the color $c(B) = c(D)$, the color $c(v)$ and possibly six more colors concerning the other three neighbors of C and oddness in their respective neighborhoods within $G - A$). So there is a color available for C. Once C is assigned with an available color, note that there is a color with an odd number of appearances in $N_{G-A}(D)$ (because $d_{G-A}(D) = 5$).

Finally, we color A. Since this is a 5-vertex, it suffices to choose a color for A so that the properness of the coloring and the oddness in $N_{G-A}(D)$ are preserved. As B and D are colored the same, and B is a 5-vertex, the number of forbidden colors at A is at most $8 = 1 + 1 + 3 \cdot 2$ (the color $c(B) = c(D)$, possibly a second color in regard to unique oddness in $N_{G-A}(D)$, and six more colors concerning the other three neighbors of A and oddness in their respective neighborhoods within $G - A$). We conclude that c indeed extends to a nice coloring of G, a contradiction. □

From Claims 9–11 it follows that the vertex v receives charge at least 1 during the discharging process, hence it cannot remain to be negatively charged. This contradiction concludes our proof.

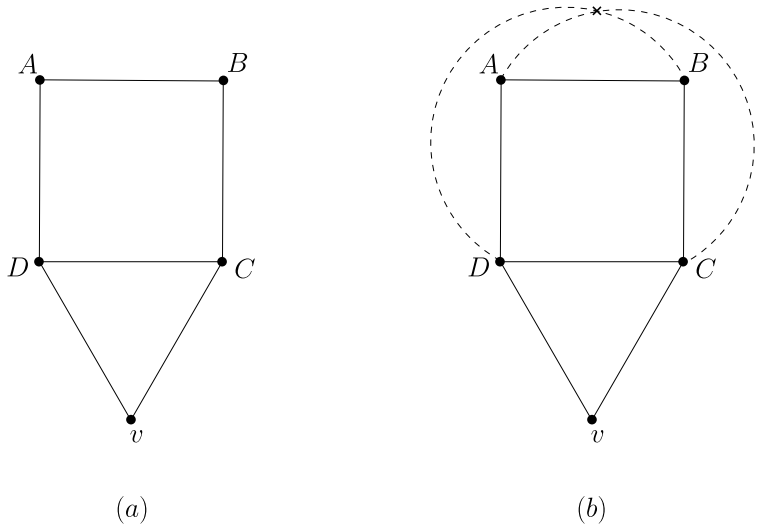

Fig. 3. A convenient face next to the 5-vertex v that happens to be a quadrangle $ABCD$, where A, B are 5-vertices and C, D are 6-vertices.

3. Further work

By Proposition 1.2, there exist plenty of planar graphs with girth 5 that require five colors for an odd coloring. We are tempted to propose the following.

Conjecture 3.1. *Every planar graph G of girth at least 6 has $\chi_o(G) \leq 4$.*

If true, Conjecture 3.1 would imply the Four Color Theorem (4CT). Indeed, say G is an arbitrary planar graph. Subdivide every edge of G once, i.e., consider the complete subdivision $S(G)$. As girth $g(S(G)) \geq 6$, take an odd 4-coloring c of $S(G)$. The restriction of c to $V(G)$ is a proper 4-coloring of G.

Although the problem of odd coloring is intriguing already in the class of plane graphs, the concept can be naturally studied also for graphs embedded in higher surfaces. For a surface Σ, we define the *odd chromatic number* of Σ,

$$\chi_o(\Sigma) = \max_{G \hookrightarrow \Sigma} \chi_o(G),$$

as the maximum of $\chi_o(G)$ over all graphs G embedded into Σ. Conjecture 1.1 and Proposition 1.2 combined purport that $\chi_o(S_0) = 5$, where S_0 is the sphere. It would be interesting to have a similar characterization to the Heawood number for other surfaces of higher genus.

Problem 3.2. Determine $\chi_o(\Sigma)$ for surfaces Σ of higher genus.

Possibly the odd chromatic number of higher surfaces coincides with the corresponding Heawood number.

Acknowledgments

This work is partially supported by ARRS, Slovenia Program P1-0383 and ARRS, Slovenia Project J1-3002.

References

[1] K. Appel, W. Haken, The solution of the four-color map problem, Sci. Amer. 237 (1977) 108–121.
[2] R. Atanasov, M. Petruševski, R. Škrekovski, Odd edge-colorability of subcubic graphs, Ars Math. Contemp. 10 (2016) 359–370.
[3] J.A. Bondy, U.S.R. Murty, Graph Theory, Graduate Texts in Mathematics, Springer, New York, 2008, p. 244.
[4] D.P. Bunde, K. Milans, D.B. West, H. Wu, Parity and strong parity edge-coloring of graphs, Congr. Numer. 187 (2007) 193–213.
[5] P. Cheilaris, Conflict-Free Coloring (Ph.D. thesis), City University of New York, 2009.
[6] P. Cheilaris, B. Keszegh, D. Pálvölgyi, Unique-maximum and conflict-free colouring for hypergraphs and tree graphs, SIAM J. Discrete Math. 27 (2013) 1775–1787.
[7] P. Cheilaris, G. Tóth, Graph unique-maximum and conflict-free coloring, J. Discrete Algorithms 9 (2011) 241–251.
[8] G. Even, Z. Lotker, D. Ron, S. Smorodinsky, Conflict-free colorings of simple geometric regions with applications to frequency assignment in cellular networks, SIAM J. Comput. 33 (2003) 94–136.
[9] I. Fabrici, F. Göring, Unique-maximum coloring of plane graphs, Dicussiones Math. Graph Theory 36 (2016) 95–102.
[10] R. Glebov, T. Szabó, G. Tardos, Conflict-free colorings of graphs, Combin. Probab. Comput. 23 (3) (2014) 434–448.
[11] M. Kano, G.Y. Katona, K. Varga, Decomposition of a graph into two disjoint odd subgraphs, Graphs Combin. 34 (6) (2018) 1581–1588.

[12] A.V. Kostochka, M. Kumbhat, T. Łuczak, Conflict-free colorings of uniform hypergraphs with few edges, Combin. Probab. Comput. 21 (2012) 611–622.
[13] B. Lužar, M. Petruševski, R. Škrekovski, Odd edge coloring of graphs, Ars Math. Contemp. 9 (2015) 277–287.
[14] B. Lužar, M. Petruševski, R. Škrekovski, On vertex-parity edge-colorings, J. Combin. Optim. 35 (2) (2018) 373–388.
[15] J. Pach, G. Tardos, Conflict-free colorings of graphs and hypergraphs, Combin. Probab. Comput. 18 (2009) 819–834.
[16] M. Petruševski, A note on weak odd edge-colorings of graphs, Adv. Math. Sci. J. 4 (1) (2015) 7–10.
[17] M. Petruševski, Odd 4-edge-colorability of graphs, J. Graph Theory 87 (4) (2018) 460–474.
[18] M. Petruševski, R. Škrekovski, Odd decompositions and coverings of graphs, Euro. J. Combin. 91 (2021) 103225.
[19] N. Robertson, D.P. Sanders, P.D. Seymour, R. Thomas, A new proof of the four colour theorem, Electron. Res. Announc. Amer. Math. Soc. 2 (1996) 17–25.
[20] S. Smorodinsky, Conflict-free coloring and its applications, in: I. Bárány, K.J. Böröczky, G.F. Tóth, J. Pach (Eds.), Geometry-Intuitive, Discrete and Convex, in: Bolyai Society Mathematical Studies, vol. 24, Springer, Berlin, Heidelberg, 2013.